Healthcare Operations & Revenue Risk Intelligence Platform

Project: Hospital Analytics Capstone project
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1. Executive Summary

Hospitals face operational inefficiencies, delayed patient care, and revenue leakage due to insurance claim rejections. This project delivers a Hospital Operations & revenue Risk Intelligence Platform, integrating SQL analytics, predictive models, and deployment-ready API. The platform helps hospital leadership optimize patient flow, allocate resources efficiently, and reduce financial losses.

The platform helps hospital leadership:

- Improved patient risk management and departmental staffing efficiency
- Reduced claim rejections and faster revenue realization
- Use a scalable, monitored, and governed AI system for predictions

Expected benefits include more efficient hospital operations, improved patient outcomes, and better financial management.

2. Business Requirements

ID	Requirement	Priority	Notes
BR-01	Monitor department visit volume	High	SQL views provide daily/weekly insights
BR-02	Identify high-risk visits	High	Risk model predicts Low/Medium/High visits
BR-03	Predict claim outcomes	High	Claim model predicts Paid/Pending/Rejected claims
BR-04	Provide audit trail and governance	Medium	Prediction logs and drift monitoring
BR-05	Enable scalable API integration	Medium	FastAPI endpoints for real-time predictions

3. System Architecture & Data Flow

Data Sources:

- CSV files containing patients, visits, and billing information

SQL Analytics Layer:

- Relational database with tables for patients, visits, and billing
- Indexed on visit_date, department, insurance_provider, claim_status

Feature Engineering:

- Visit frequency per patient
- Average length of stay be department
- Provider rejection rate
- Days since registration

Predictive Models:

- Risk Model: Random Forest predicting visit risk (Low, Medium, High)
- Claim Model: Random Forest predicting claim outcomes (Paid, Pending, Rejected)

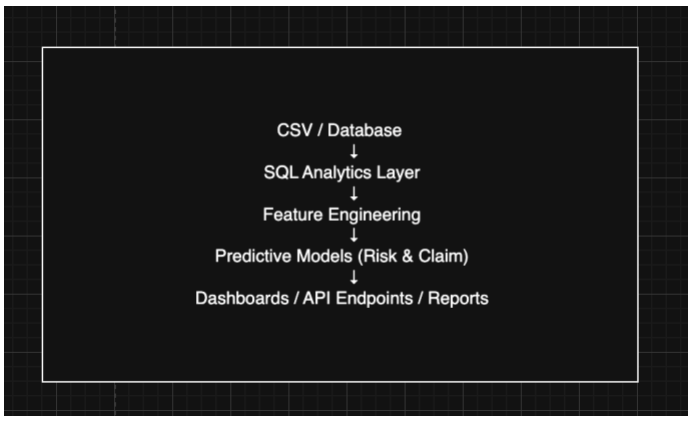
Deployment:

- FastAPI service with endpoints /predict_risk and /predict_claim
- Containerized using Docker for reproducibility and scalability

Monitoring and Governance:

- Data validation checks, feature drift detection, prediction logging, and retraining strategy

Data Flow:



4. Key Operational Insights

Operational Metrics:

Metric	Finding
Top departments by volume	ER: 3500, OPD: 2800, ICU: 1200
Highest avg length of stay	ICU: 72h, ER: 48h
% High-risk visits by dept	ICU: 32%, ER: 20%
Avg visits per patient by city	City A: 2.3, City B: 1.8

Financial Insights:

Metric	Finding
Top insurance providers by billed amount	Provider X: $2.3M, Provider Y: $1.8M
Avg payment delay	12 days
Revenue realization ratio	ER: 0.87, ICU: 0.91
Visits billed but not approved	23 flagged cases

5. Model Performance & Business Impact

Model	Key Metric	Business Implication
Risk Model	Recall (High Risk) 0.85	Helps staff proactively allocate resources in ER/ICU
Claim Model	Recall (Rejected) 0.78	Reduces revenue losses and improves cash flow

Impact Estimate:

- Correctly predicting 80% of high-risk visits allows ~10% more efficient staffing which leads to estimated savings $50,000/month

- Predicting rejected claims early can improve cash flow by ~$75,000 per quarter

6. Deployment & Scaling Strategy

- FastAPI endpoints provide real-time predicitons
- Docker returns reproducible and scalable deployment
- System can scale to additional hospitals and departments
- Drift monitoring triggers retraining if feature distributions change significantly
- Retraining is scheduled quarterly or when drift is detected

7. Governance & Compliance

To ensure that the Hospital Operations & Revenue Risk Intelligence Platform operates safely, reliably and in line with industry standards, the governance and compliance framework has been aligned with the NIST AI Risk Management Framework (AI RMF). This approach provides a structured guidance on managing operational, financial, and data-related risks while promoting transparency, accountability, and ongoing monitoring of AI models.

NIST AI RMF Function	Application in This Project
Govern	We have policies and procedures for how models are used, including versioning, audit logs, retraining schedules, and clear documentation of assumptions and limitations.
Map	We identify operational and financial risks such as high-risk patient visits, claim rejections, revenue leakage, and potential data quality issues.
Measure	We monitor model performance metrics (like precision, recall, F1-score) and business KPIs such as revenue realization and patient flow efficiency.
Manage	Validation checks ensure inputs are complete and valid, features have known categories, and feature/prediction drift is detected and addressed.
Protect	Patient and financial data are handled securely, audit logs are anonymized, and API access is controlled to prevent unauthorized use.

Implementation Notes:

- Audit logs record each prediction along with the corresponding model version and outcome, supporting compliance reviews and traceability.
- Drift detection and scheduled retraining help maintain consistent, reliable model performance over time.
- Input validation flags unknown or invalid data, reducing operational and financial risks.
- Overall, the system aligns with NIST guidance for trustworthy AI by emphasizing transparency, accountability, and continuous monitoring.

8. Business Recommendations

- Start deployment in ER and ICU for maximum impact
- Use claim model predictions to follow up on high-risk or rejected claims proactively
- Review drift reports and retrain models quarterly
- Integrate SQL dashboards into daily operations for actionable insights
- Expand predictive capabilities to other departments after initial deployment